

END-SEMESTER REVIEW · NOVEMBER 2026

LabLend — Lab Equipment Borrowing and Return Tracking System

Replacing the lab logbook with a live borrow-and-return workflow

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CB23F35 – Low-Code Application Development using Zoho Creator

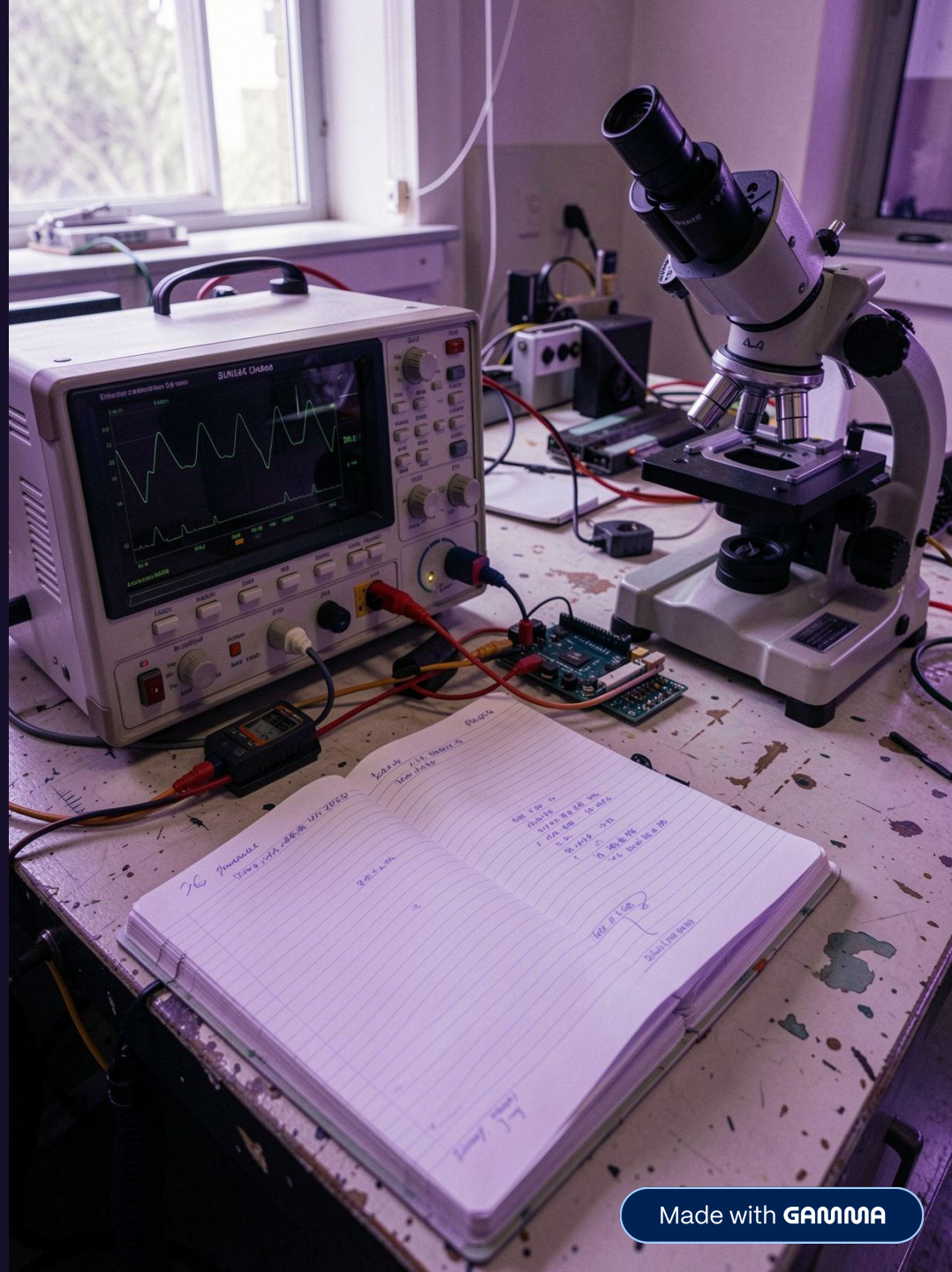
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01 — THE PROBLEM

Paper logs hide the real state of the lab

The record exists, but the workflow does not.

- Equipment is lent through a paper logbook: oscilloscopes, multimeters, Arduino, Raspberry Pi, microscopes
- No real-time availability; students must walk to the lab to check
- Approvals and return dates are verbal, so nothing is recorded
- Overdue items surface only during stock checks
- Return condition is missing, and one item can be promised twice



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Objectives and scope define a focused first release

A small surface area, with the state changes made explicit.



Linked forms

Four linked forms with automated request lifecycle



Live status

Equipment status stays synchronized automatically



Visibility

Overdue flag and live dashboard for lab staff

In scope

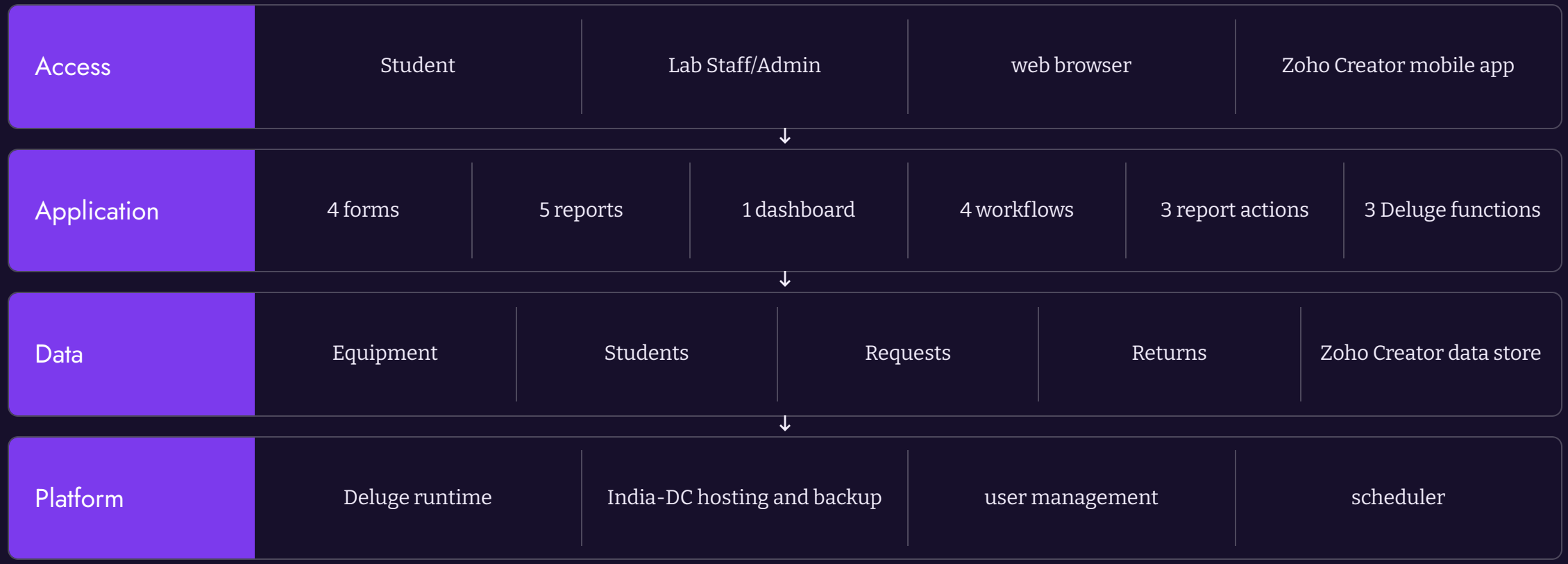
Web and mobile app on Zoho Creator

Out of scope

Fines, email/SMS, college SSO, QR scanning

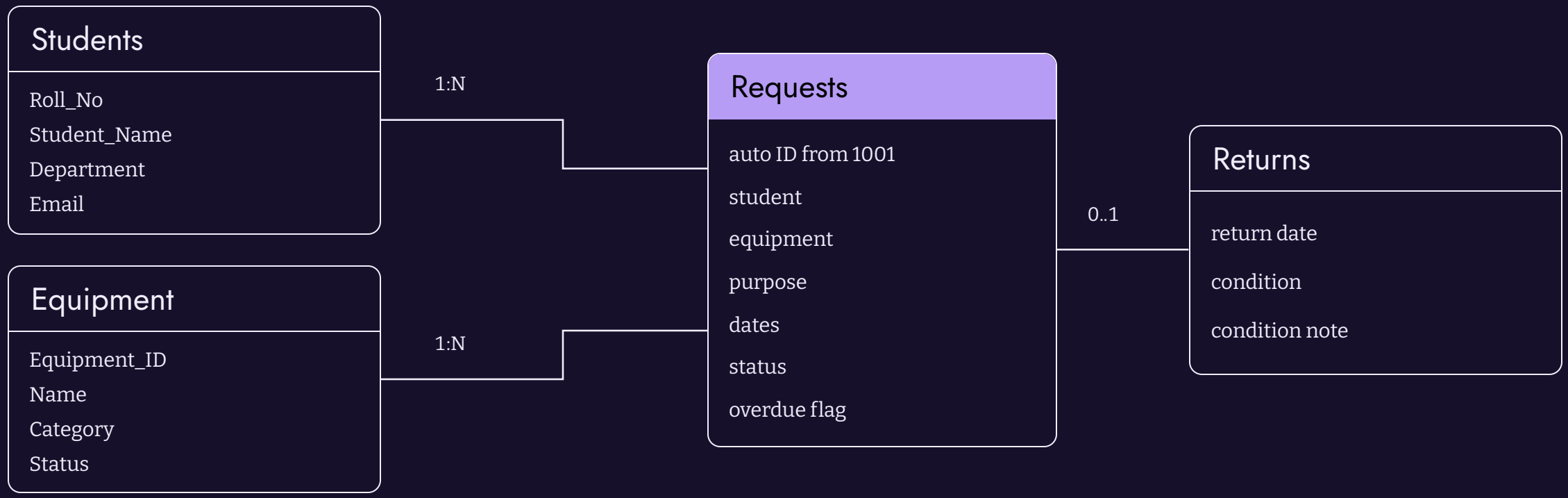
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Four layers connect access, logic, data, and platform services



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The data model makes every loan traceable



Students and equipment can have many requests; each request has zero or one return record.

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Implementation separates configuration from custom logic

Forms

Equipment, Students, Requests, Returns with lookups, drop-downs, mandatory fields

Reports

Equipment, students, requests, overdue items, returns; status-aware actions

Workflows

Request defaults, date validation, return defaults, return processing

Deluge

approveRequest, rejectRequest, issueRequest run from conditional report buttons



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The request lifecycle blocks bad states before they spread

Validation happens at entry, transition, issue, and return.

On Load

Request_Date to today; Status locked Pending

Available only

Only Available equipment appears in the drop-down

On Validate

Blocks invalid dates or equipment no longer Available

Status-gated

Approve/Reject for Pending; Issue for Approved

Return

Updates request, equipment status, late flag, and repair state

Pending → Approved / Rejected → Issued → Returned

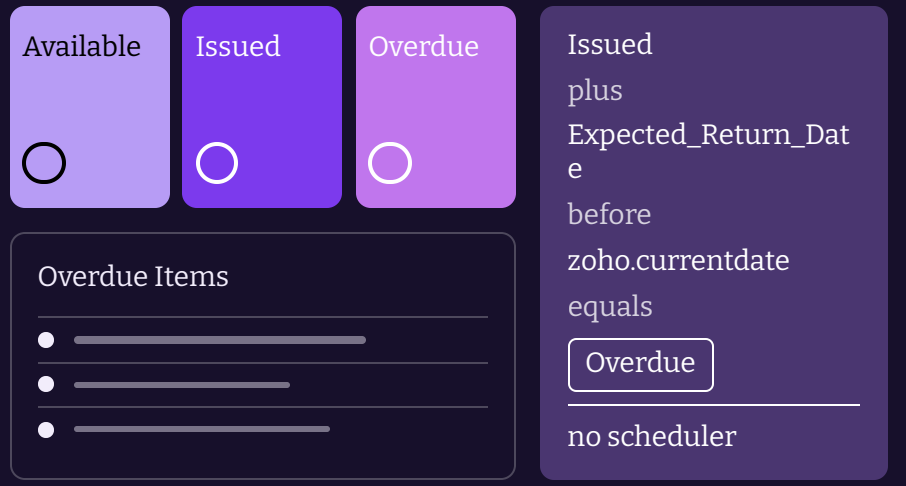
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07 — LIVE OVERDUE CONTROL

Overdue detection is live, visible, and scheduler-free

A calculated condition keeps the dashboard current without a nightly job.

- **Rule:** Status == “Issued” AND Expected_Return_Date < zoho.currentdate
- Calculated in the Overdue Items report and dashboard; no nightly job needed
- Dashboard KPIs: Available in green, Issued in blue, Overdue in red, plus the overdue list
- Issue logic checks approval and availability before changing both request and equipment to Issued




```
if(request.Status == "Approved"
&& request.Equipment.Status == "Available")
{
  request.Status = "Issued";
  request.Equipment.Status = "Issued";
}
```


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Testing confirms the workflow and exposes the next gaps

14

test cases executed; all passed

DASHBOARD DEMO

5/0/0 → 3/2/1 → 4/1/1

After issuing 2 requests, then 1 return

EDGE CASES

1002 overdue

Double booking blocked by validation

LIMITATIONS

Free plan disables schedules and environments; separate student/staff permission profiles are not yet configured

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09 — CLOSE

LabLend turns a fragile logbook into a traceable workflow

Complete lending flow built by one developer on low-code

Deluge is reserved for the points configuration cannot handle

NEXT

Permission profiles, daily email reminders, QR scanning, calendar view, usage analytics

Thank You!

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